

Monetary policy and endogenous financial crises

F. Boissay, F. Collard, J. Galì, C. Manea

Giacomo Cattelan

BGGM

December 8 2021

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Should monetary policy take financial fragility into account?

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Neo-Keynesian model augmented with

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- ▶ endogenous capital accumulation → the economy deviates persistently from its steady state;
- ▶ firms are subject to idiosyncratic shocks → give rise to a loan market among firms;
- ▶ financial frictions on the loan market: private information and imperfect commitment.

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- ▶ Provide an explanation for endogenous financial crisis;
- ▶ monetary policy effect on an economy with financial frictions;
- ▶ optimal monetary policy: Strict Inflation Targeting (STI), vs Standard Taylor Rule (STR) vs Interest Rate Rules (IR).

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- ▶ firms produce intermediate goods and sell it competitively to retailers;
- ▶ Central Bank setting interest rate according to a Taylor rule:

$$1 + i_t = \frac{1}{\beta}(1 + \pi_t)^{\phi_\pi} \left(\frac{Y_t}{Y} \right)^{\phi_y}$$

Household

$$\max_{\{C_t, N_t\}_t} \mathbb{E}_0 \sum_{t \geq 0} \beta^t \left(\frac{C_t^{1-\sigma}}{1-\sigma} - \vartheta \frac{N_t^{1+\nu}}{1+\nu} \right)$$

s.t.

$$P_t C_t + B_{t+1} + P_t K_{t+1} \leq P_t \omega_t N_t + \frac{1}{Z_{t-1}} (1 + i_{t-1}) B_t + P_t (1 + r_t^k) K_t + \mathcal{X}_t$$

$$C_t = \left(\int_0^1 C_t(j)^{\frac{\epsilon-1}{\epsilon}} dj \right)^{\frac{\epsilon}{\epsilon-1}} \text{ and } P_t = \left(\int_0^1 P_t(j)^{\epsilon-1} dj \right)^{\frac{1}{\epsilon-1}}$$

► FOC

Retailers

$$\max_{\{P_t(j)\}_t} \mathbb{E}_0 \sum_{t \geq 0} \Lambda_{0,t} \left(\frac{P_t(j)}{P_t} Y_t(j) - \frac{(1-\tau)p_t}{P_t} Y_t(j) - \frac{\varrho}{2} Y_t \left(\frac{P_t(j)}{P_{t-1}(j)} - 1 \right)^2 \right)$$

s.t.

$$Y_t(j) = \left(\frac{P_t(j)}{P_t} \right)^{-\epsilon} Y_t$$

$$Y_t = \left(\int_0^1 Y_t(j)^{\frac{\epsilon-1}{\epsilon}} dj \right)^{\frac{\epsilon}{\epsilon-1}} \text{ and } \Lambda_{t,t+k} = \beta^k \left(\frac{C_{t+k}}{C_t} \right)^{-\sigma}. \quad \text{FOC}$$

The average markup is given by $\mathcal{M}_t = \frac{P_t}{(1-\tau)p_t}$

Firms

Produce intermediate good:

$$y_t(q) = A_t(qK_t(q))^\alpha(N_t(q))^{1-\alpha} \quad q = \begin{cases} 0 & \text{w.p. } \mu \\ 1 & \text{w.p. } 1 - \mu \end{cases}$$

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- ▶ born identical at the end of $t - 1$, choose K_t issuing an amount $P_{t-1}K_t$ of equity;

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- ▶ profits are realized and rebated to equity holders, un-depreciated capital is sold.

Firms

Profits:

$$\frac{1}{(1-\tau)\mathcal{M}_t} y_t(q) - \omega_t N_t(q) + (1-\delta)K_t(q) - (1+r_t^\ell)(K_t(q) - K_t)$$

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Return on capital:

$$r_t^k(q) = \frac{1}{(1-\tau)\mathcal{M}_t} \frac{y_t(q)}{K_t} - \omega_t \frac{N_t(q)}{K_t} - (r_t^\ell + \delta) \frac{K_t(q) - K_t}{K_t} - \delta$$

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► if $q = 0$, $N_t(0) = 0$ and $K_t(0) \begin{cases} = 0 & \text{if } r_t^\ell + \delta > 0 \\ \in [0, K_t] & \text{if } r_t^\ell + \delta = 0 \end{cases}$

$$r_t^k(0) = -(r_t^\ell + \delta) \frac{K_t(0) - K_t}{K_t} - \delta = r_t^\ell$$

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- if $q = 1$, the firm hires optimally (► FOC), so, letting $\Phi_t = \frac{y_t(1)}{K_t(1)}$

$$r_t^k(1) = \frac{\alpha \Phi_t}{(1-\tau)\mathcal{M}_t} \frac{K_t(1)}{K_t} - (r_t^\ell + \delta) \frac{K_t(1) - K_t}{K_t} + r_t^\ell$$

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$$K_t(1) > K_t \iff \frac{\alpha \Phi_t}{(1-\tau)\mathcal{M}_t} - \delta > r_t^\ell$$

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- ▶ in equilibrium, $r_t^k(1) = r_t^k(0) = r_t^\ell$ and unproductive firms lend all their capital to productive ones.
- ▶ in aggregate $K_t = (1 - \mu)K_t(1)$ and $N_t = (1 - \mu)N_t(1)$, so

$$y_t \equiv (1 - \mu)y_t(1) = A_t K_t^\alpha N_t^{1-\alpha}$$

Frictional Loan Market

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Frictional Loan Market

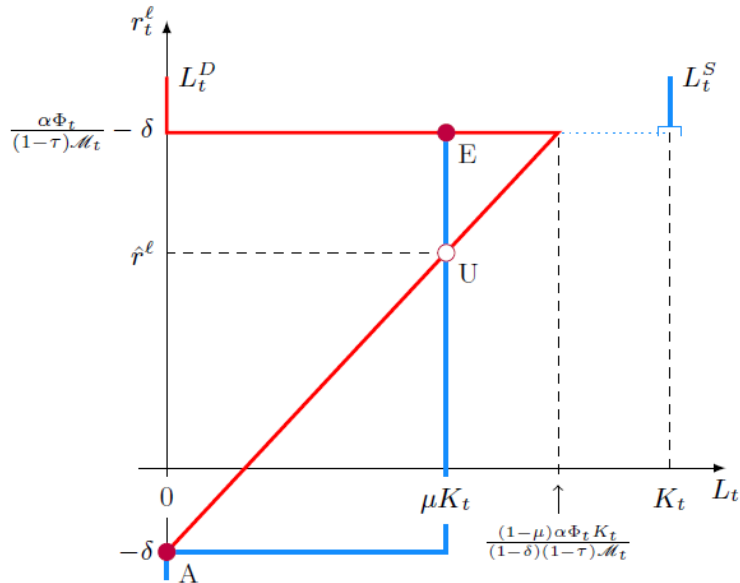
- ▶ Idiosyncratic shocks are un-observable;
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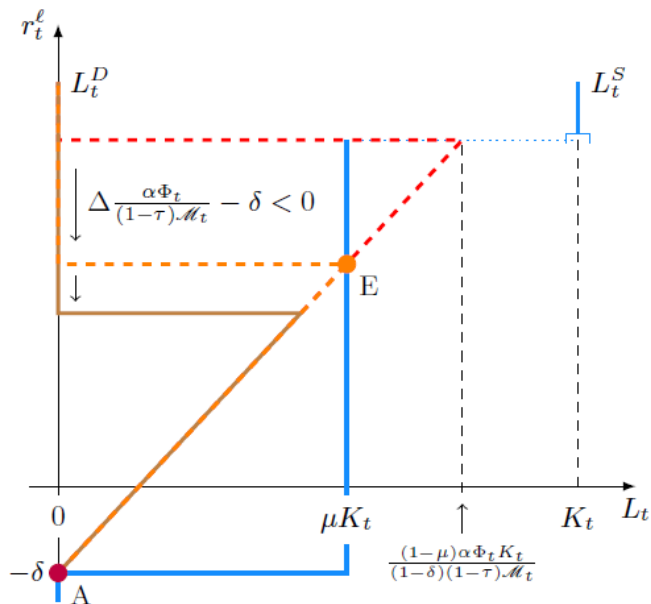
- ▶ Idiosyncratic shocks are un-observable;
- ▶ unproductive firms can choose to borrow capital, stay idle, sell it and abscond. The implied return on this operation is $P_t(1 - \delta)K_t(1)$;
- ▶ incentive compatibility constraint: lending capital to productive firms, must be more profitable than defaulting:

$$P_t(r_t^\ell + \delta)K_t \geq P_t(1 - \delta)K_t(1) \iff \frac{K_t(1) - K_t}{K_t} \leq \psi_t = \frac{r_t^\ell + \delta}{1 - \delta}$$

Loan Market Equilibrium



Financial Crisis



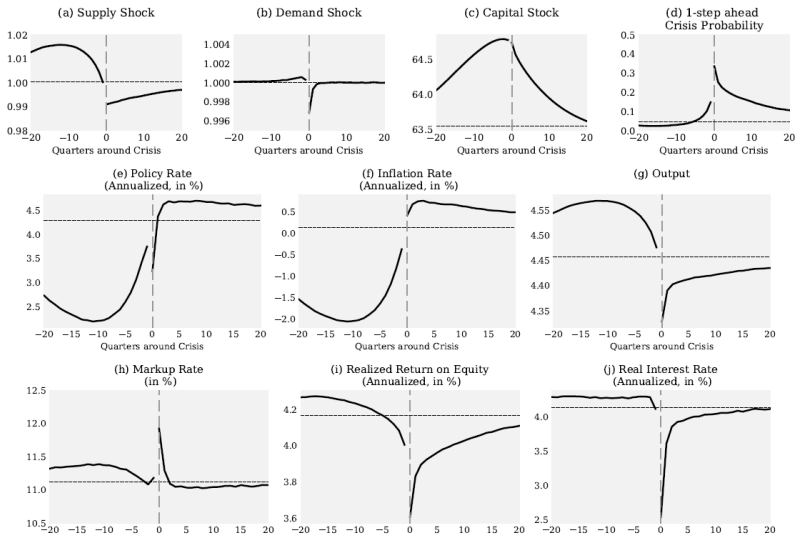
Financial Crisis

Probability of a crisis:

$$\mathbb{P}_t \left[\frac{\alpha Y_t}{(1 - \tau) \mathcal{M}_t K_t} < \hat{r}_t^\ell + \delta \right]$$

- ▶ fall in aggregate output (Y-channel);
- ▶ rise in markup (M-channel);
- ▶ high capital accumulation (CA-channel).

Crisis Simulation



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Absent financial frictions, SIT ($\phi_\pi = \infty$ and $\phi_y = 0$) is the first best.

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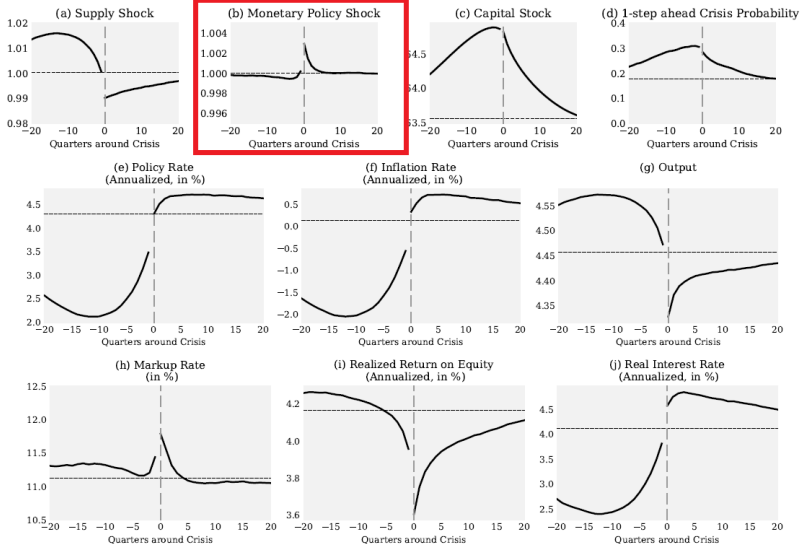
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- ▶ agents foresee financial crisis, save more inducing capital accumulation and triggering the crisis itself: this can be prevented.

Is there welfare gain from a "leaning against the wind" policy ($\phi_y > 0$)?

Welfare Gains from Leaning Against the Wind

	Rule paramaters		Frictionless economy	Frictional economy							
	ϕ_π	ϕ_y	PCE	PCE	Crisis	Length	Output	Y-, M-, and CA-channels			
			(%)	(%)	time (%)	(quarters)	loss (%)	$\sigma(Y_t)$	$\sigma(\mathcal{M}_t)$	$\sigma(K_t)$	$\rho(Y_t, \mathcal{M}_t)$
SIT	$+\infty$	0.000	—	—	5.03	4.59	-5.60	3.92	0.00	3.70	0.00
STR	1.500	0.125	-0.0062	-0.0408	[8.00]	1.78	-3.20	3.62	1.09	3.16	0.14
IR-[0.212]	1.500	0.212	-0.0059	-0.0116	[5.03]	1.78	-2.83	3.26	1.06	2.77	0.50
IR-[0.309]	1.500	0.309	-0.0075	0.0117	[2.50]	1.68	-2.54	2.93	1.16	2.42	0.72
IR-[0.415]	1.500	0.415	-0.0101	0.0239	[1.00]	1.54	-2.26	2.67	1.32	2.11	0.84
IR-[0.491]	1.500	0.491	-0.0124	0.0267	[0.50]	1.47	-2.12	2.50	1.43	1.92	0.88

Discretionary Monetary Policy



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- ▶ non-linearities which trigger endogenous large crisis from small shocks;
- ▶ non-trivial role for monetary policy:
 - ▶ it corrects financial distortions if rule-based;
 - ▶ it can foster exuberance and increase crisis probability if discretionary.

$$\beta \mathbb{E}_t \left[\left(\frac{C_{t+1}}{C_t} \right)^{-\sigma} (1 + r_{t+1}) \right] = Z_t$$

$$\beta \mathbb{E}_t \left[\left(\frac{C_{t+1}}{C_t} \right)^{-\sigma} (1 + r_{t+1}^k) \right] = 1$$

$$\vartheta N_t^\nu C_t^\sigma = \omega_t$$

▶ Back

$$\omega_t = \frac{1 - \alpha}{(1 - \tau)\mathcal{M}_t} \frac{y_t(1)}{N_t(1)}$$

$$\frac{y_t(1)}{K_t(1)} = \Phi_t = A_t^{\frac{1}{\alpha}} \left(\frac{1 - \alpha}{(1 - \tau)\mathcal{M}_t\omega_t} \right)^{\frac{1-\alpha}{\alpha}}$$

▶ Back